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Conservation Status, Population Trends, and Management Recommendations for *Aviofauna militaris* Taxa: An IUCN Red List Assessment Framework

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ABSTRACT
We present the first formal conservation assessment for the clade Aviofauna militaris using adapted IUCN Red List criteria. Of the twelve recognized species, three qualify as Critically Endangered (B-2 Spirit, F-117 Nighthawk, and one undescribed Stealthia taxon), four as Endangered, three as Vulnerable, and two as Near Threatened. Primary threats include direct mortality from combat operations, habitat loss at nesting and roosting sites, reduced reproductive success due to disturbance, and potential genetic bottlenecks in small, isolated populations. We provide species-specific management recommendations, including establishment of protected nesting zones, restrictions on operations during breeding seasons, and development of non-invasive population monitoring protocols. The assessment highlights the urgent need for international cooperation in the conservation of these unique technobiological taxa.

Keywords: Aviofauna militaris, IUCN Red List, conservation status, population trends, threat assessment, military ornithology, technobiological conservation

1. INTRODUCTION
The recognition that *Aviofauna militaris* taxa are living organisms with finite populations, reproductive strategies, and ecological requirements necessitates the application of conservation biology frameworks to their management. This paper presents the first systematic assessment of conservation status using criteria adapted from the IUCN Red List Categories and Criteria.

2. METHODS
Assessment followed the IUCN Red List Categories and Criteria (Version 3.1) with modifications appropriate for technobiological taxa as outlined in the 2025 Guidelines for the Assessment of Cryptobiological Taxa. Population data were derived from a combination of direct counts at known aggregation sites, mark-recapture studies using non-invasive identification methods, and demographic modeling based on observed reproductive rates.

3. RESULTS
3.1 Critically Endangered Taxa
B-2 Spirit is assessed as Critically Endangered under criteria A2cd and C2a(i). Global population is estimated at fewer than 20 mature individuals. Primary threats include extremely low reproductive rate, high sensitivity to disturbance at roosting sites, and ongoing operational losses.

3.2 Endangered and Vulnerable Taxa
Four species meet Endangered criteria and three meet Vulnerable criteria. Common threats across these taxa include habitat degradation at nesting sites, direct mortality from training and combat operations, and potential inbreeding depression in small, fragmented populations.

3.3 Management Recommendations
We recommend: (1) establishment of seasonal exclusion zones around known nesting colonies during breeding periods; (2) development of non-invasive population monitoring using acoustic and satellite methods; (3) creation of artificial roosting structures for troglodytic taxa; (4) international agreements limiting operations that result in high adult mortality during breeding seasons; and (5) ex-situ conservation programs for the most critically endangered taxa.

***Figure 1.** Population trajectory projections for B-2 Spirit under current management and under recommended conservation interventions.*

4. DISCUSSION
The conservation status of *Aviofauna militaris* presents unique ethical and practical challenges. Unlike natural species, these taxa are simultaneously biological organisms and strategic military assets. Effective conservation will require unprecedented cooperation between military authorities, conservation agencies, and international bodies. The window for meaningful intervention is narrow, particularly for the most threatened taxa.

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